Thirsty Engines: Why Your 1974 Car Hates Your Wallet

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The question

- Humanity's oldest question: *why am I at the gas station again?*
- Data: 32 cars road-tested by *Motor Trend* in 1974
- **Today:** do cylinders and horsepower make cars thirsty?

Puzzle

- The Toyota Corolla gets **33.9** mpg...
- ...the Cadillac Fleetwood gets **10.4** mpg
- What is the Cadillac doing with all that gas?

Argument

1. **Many Mouths:** each cylinder is a tiny mouth that drinks gasoline
2. **Enthusiasm:** horsepower is *how hard* those mouths drink
3. Drawing on Arel-Bundock (2025) to communicate our estimand more effectively

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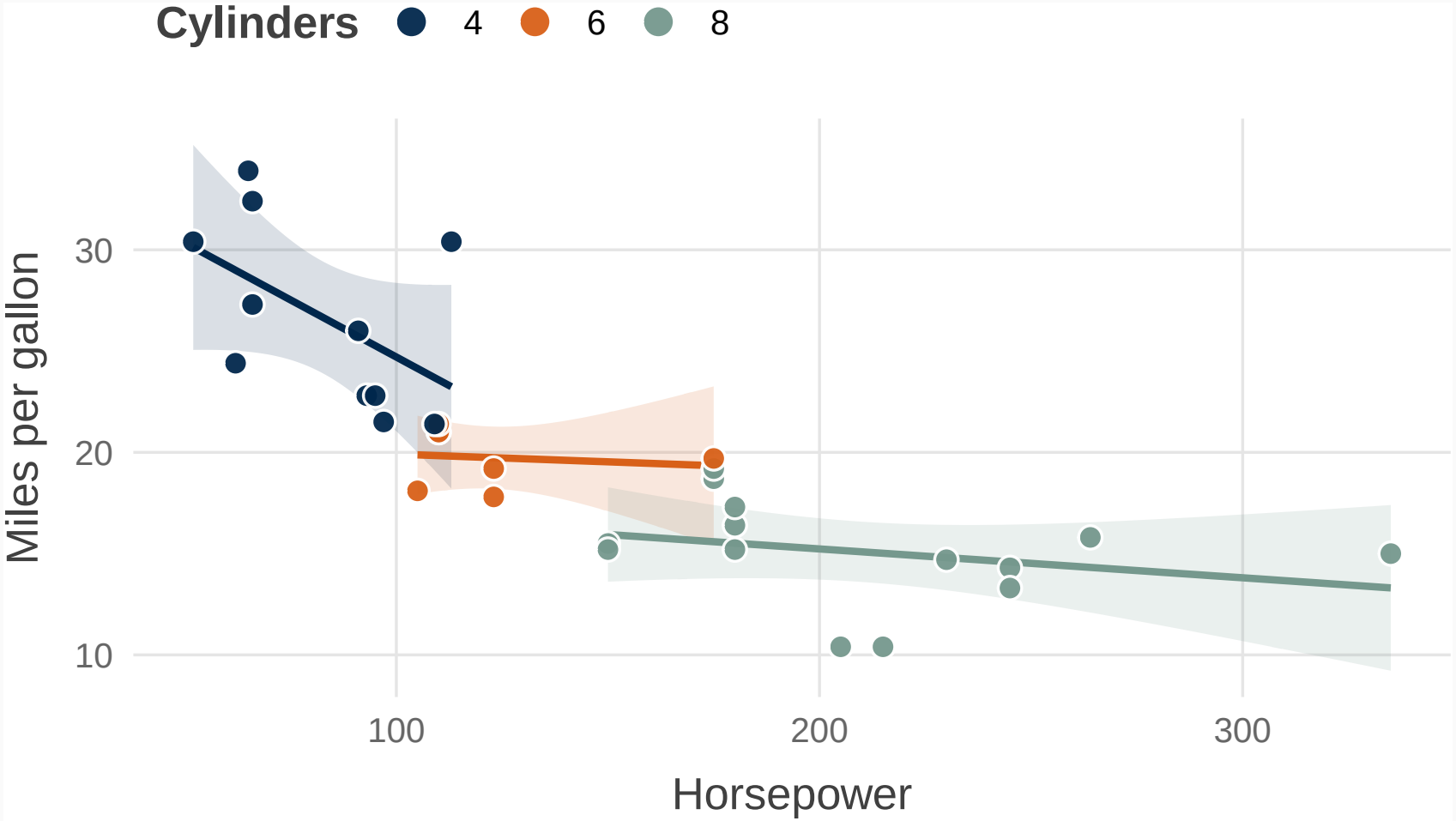
H1: More cylinders $\rightarrow$ fewer miles per gallon

H2: More horsepower $\rightarrow$ fewer miles per gallon

Empirical strategy

$$\text{MPG}_i = \beta_0 + \beta_1 \text{Cylinders}_i + \beta_2 \text{Horsepower}_i + \varepsilon_i$$

- `mtcars` ($N = 32$), the most-analyzed dataset in human history
- OLS, because I know how to do it
- ε_i: aerodynamics, driver mood, Led Zeppelin on the 8-track


Cylinders
4
6
8
Miles per gallon
30
20
10
100
200
300
Horsepower

The determinants of engine thirst

	Model 1
Constant	28.650*** (1.588)
6 cylinders	-5.968** (1.639)
8 cylinders	-8.521** (2.326)
Horsepower	-0.024 (0.015)
Num.Obs.	32
R2	0.754

Takeaways

- **H1 confirmed:** an 8-cylinder car loses ≈ 8.5 mpg vs. a 4-cylinder
- **H2 runs out of gas:** horsepower is guilty in spirit, but $p \approx 0.13$
- Next steps: data from after 1974, and asking who *needs* a V8 for groceries

References

Arel-Bundock, Vincent. 2025. *Model to Meaning: How to Interpret Statistical Models with r and Python*. 1st ed. Chapman; Hall/CRC. https://doi.org/10.1201/9781003560333.